A FUNCTOR ADMITTING A REPRESENTABLE MORPHISM TO AN ALGEBRAIC SPACE IS AN ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement carries the author's slogan environment; it is part of the counted statement span and is preserved verbatim, with the slogan declaration retained in the preamble.

Editorial note. The six references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Editorial note. Revision r002 only changes the preamble span: the retained preamble was extended with \usepackage{verbatim}, restored verbatim from the chapter's own preamble.tex line 8, because the counted statement uses the author's slogan environment which is defined through \comment. No statement or proof text changed; r001 failed on the undefined \comment.

Allowed standard prerequisites (editorial).

- A representable transformation with sheaf target has sheaf source. Lemma lemma-representable-transformation-to-sheaf (433-439), used at proof lines 1939-1940. Gives the sheaf condition for G.
- Base change of a representable transformation is representable. Lemma lemma-base-change-representable-transformations, used at proof lines 1947-1949. Makes G x_F G -> G x G representable.
- The diagonal of a representable transformation is representable. Lemma lemma-representable-transformation-diagonal (456-462, produced as L44186), used at proof lines 1950-1952. Gives representability of G -> G x_F G.
- Composition of representable transformations. Lemma lemma-composition-representable-transformations, used at proof lines 1953-1955. Makes Delta_G representable.
- Base change keeps the property of being surjective and etale. Lemma lemma-base-change-representable-transformations-property, used at proof lines 1958-1960. Produces the surjective etale chart U' -> G.

Source reference. Original definition 025Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2831–2831. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3798–3798. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3808–3808. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3797–3797. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05L9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7228–7228. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3800–3800. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme contained in Sch_{fppf}. Let F be an algebraic space over S. Let $G \to F$ be a representable transformation of functors. Then G is an algebraic space.*

Proof. By Lemma 02WD we see that G is a sheaf. The diagram

$$\begin{array}{ccc} G \times_F G & \longrightarrow & F \\ \downarrow & & \downarrow{\scriptstyle \Delta_F} \\ G \times G & \longrightarrow & F \times F \end{array}$$

is cartesian. Hence we see that $G \times_F G \to G \times G$ is representable by Lemma 02WB. By Lemma 05L9 we see that $G \to G \times_F G$ is representable. Hence $\Delta_G : G \to G \times G$ is representable as a composition of representable transformations, see Lemma 02WA. Finally, let U be an object of $(Sch/S)_{fppf}$ and let $U \to F$ be surjective and étale. By assumption $U \times_F G$ is representable by a scheme U'. By Lemma 02WL the morphism $U' \to G$ is surjective and étale. This verifies the final condition of Definition 025Y and we win. □